Program and Branch Name: B.Tech CSE

PBL PROGRESS REPORT – II

Course Name	Computer Architecture and Parallel Processing (CAPP)
Course Code	CCSE0304
Faculty Name	Ms. Pooja Kumari
Assignment Type	Group Assignment
Student(s) Name	Deepak Joshi, Aman Agrawal, Manish Kumar
Roll No. / ERP ID	2501331720045, 2501331540037, 2501331720069
Team/Group No. (Group/Capstone only)	52
Problem Statement / Project Title	Smart Traffic Processing System
SDG(s)	SDG 11 – Sustainable Cities and Communities
Reporting Month	Month 2
Date of Submission	17/10/2026

Student Declaration

I/We certify that this assignment is my/our own work, with all sources appropriately acknowledged. I/We understand that 0% mark and academic penalties may apply if content is plagiarised or inappropriately AI-generated.

Student Name & Signature(s):

Deepak Joshi, Aman Agrawal, Manish Kumar







Date: 17-10-2026

PBL PROGRESS REPORT – II

Review 2: Development & Application

Review Coverage: Individual / Group Assignment: Unit/Module 3 (Control Unit, Instruction Cycle, RISC/CISC, Pipelining, Microprogramming, Flynn's Classification) and Unit/Module 4 (Parallel Architectures, Memory Hierarchy, Cache Coherence Protocols)

Field	Details
1. Previous Review Feedback	In Review 1 the problem statement and scope were finalised. The next step was to show how the processor handles the traffic data and to link each topic to a part of our system.
2. Action Taken on Previous Feedback	• Designed the instruction format, instruction set and instruction cycle for the traffic calculations • Linked each Unit 3 and Unit 4 topic to a part of the system (see Field 4)
3. Research / Data Collection Progress	Studied Unit 3 and Unit 4 topics: instruction formats, instruction cycle, RISC vs CISC, pipeline hazards, Flynn's classification, cache memory and the MESI protocol. We do not have live camera data yet, so sample values of vehicle count and queue length for each junction were assumed.
4. Unit/Module Concepts Applied	• **Unit 3:** instruction formats, instruction cycle, pipelining, hardwired and microprogrammed control, Flynn's classification • **Unit 4:** multi-core (shared memory) architecture, cache memory, MESI cache coherence protocol (Modified, Exclusive, Shared, Invalid), Amdahl's law
5. Methodology / Approach Followed	We followed a step by step approach: (1) decide the instructions needed to calculate traffic load, (2) design the instruction cycle, (3) pipeline the execution stages, (4) use a multi-core MIMD design where each core handles some junctions, (5) use MESI to keep shared signal data correct across cores.
6. Work Completed During Month 2	• Defined a 16-bit instruction format (I-bit, opcode, address) • Selected the instructions LOAD, ADD, MUL, STORE and BUN • Mapped fetch, decode and execute steps to one signal decision • Designed a 4-stage pipeline (Fetch, Decode, Execute, Write-back) • Selected MIMD multi-core design with MESI protocol
7. Analysis / Development / Implementation	• Calculated approach load L = (N1 + N2) x W using one, two and three address instructions (N = vehicles in a lane, W = vehicle weight) • Checked the pipeline for data hazard (ADD waiting for LOAD) and control hazard (branch when queue crosses the limit), handled using forwarding and a stall • Hardwired control chosen for the normal signal calculation because it is faster; microprogrammed control can be used where rules change, such as emergency vehicle priority • Shared data like signal state is kept consistent across cores using MESI

Field	Details
8. Results / Interim Outcomes	• Pipeline speed-up = nk / (k + n - 1). For k = 4 stages and n = 100 instructions, speed-up = 400 / 103 = 3.88 (theoretical) • Amdahl's law: speed-up = 1 / ((1 - P) + P/N). With P = 0.8 and N = 4 cores, speed-up = 2.5 • Different junctions can be processed at the same time on different cores • Design is complete; actual speed-up will be measured after simulation
9. Evidence / Supporting Material	Instruction format table, instruction cycle flowchart, pipeline timing diagram, MESI state diagram, multi-core block diagram and architecture document.
10. Challenges & Corrective Actions	**Challenge:** the decision depends on the density result, so all stages cannot run in parallel; choosing between hardwired and microprogrammed control; understanding MESI state changes. **Corrective action:** used forwarding and a stall, compared both control types, and traced MESI states with small examples.
11. Individual / Team Contribution	• Deepak Joshi (Team Leader) - instruction set design, combining Unit 3 and Unit 4, coordination • Aman Agrawal - instruction cycle, pipelining, hazards, documentation • Manish Kumar - sample data, memory hierarchy, parallel architecture, cache coherence
12. Overall Progress (%)	75% (instruction set, pipeline and coherence design done; simulation and testing remaining)
13. Plan for Final Review	• Simulate the pipelined decision path for multiple junctions • Implement and test the MESI model for shared signal data • Compare sequential and parallel execution time • Complete Unit 5 (consistency, synchronization, interconnects), final report and demo